Newborough Warren: what is happening to the dune water table

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A plain-language summary of a 21-year study, for anyone who cares about the Warren.

Why the water table matters

Newborough Warren's wildlife lives in its damp hollows — the dune slacks. Whether a hollow is a flower-rich **wet slack** or dries out to ordinary grassland comes down to a few centimetres of water table through the summer. Get that wrong and the special plants and animals go with it.

For 21 years this study has followed the Warren's water table using a network of 88 simple, hand-measured wells and freely available weather records. The aim was plain: work out what is changing, why, and what management can — and cannot — do about it. Two figures accompany this summary: one on **what coast and climate do to the water table**, and one on **what management does**.

The headline: two big pressures, both larger than anything management can do

Two forces are pulling the Warren's water table down, and both are bigger than the tools available to manage the site.

Climate — the steady one. Right across the site, the water table is drifting down year on year — as rainfall recharge falls and early summers grow hotter and thirstier. It is not dramatic in any single year, but it never stops and it acts *everywhere at once*. We can see the fall in the levels themselves. What twenty-one years cannot do is say **how fast** it is going, because the swing from one year to the next is larger than the drift hidden inside it. So we describe this pressure by its direction and its reach, and we do not put a number on its rate.

Coastal erosion — the sleeper. Where the sea is cutting into the dune front, it draws the water table down with it. This effect is **strongest right at the shore and fades inland**, reaching background levels roughly **900 metres** in. It has been easy to overlook, but it is a serious and ongoing pressure. It also works by *exactly the same physics* as one of the management tools below — a point we return to.

Where the two meet. Near the coast, erosion dominates. Further inland its effect fades, while the climate pressure acts everywhere at once — so the further from the sea you go, the more of what you see is the site-wide drift rather than the shore (Figure 1). We can no longer say where the two cross: that would need a rate for the climate term, and we do not have one.

A note on certainty: the *size* of the coastal effect is modelled, and the well network cannot pin it down on its own — but its **direction and mechanism are consistent with what has been documented at comparable dune systems in Wales**. We state the mechanism with confidence and the exact numbers with caution.

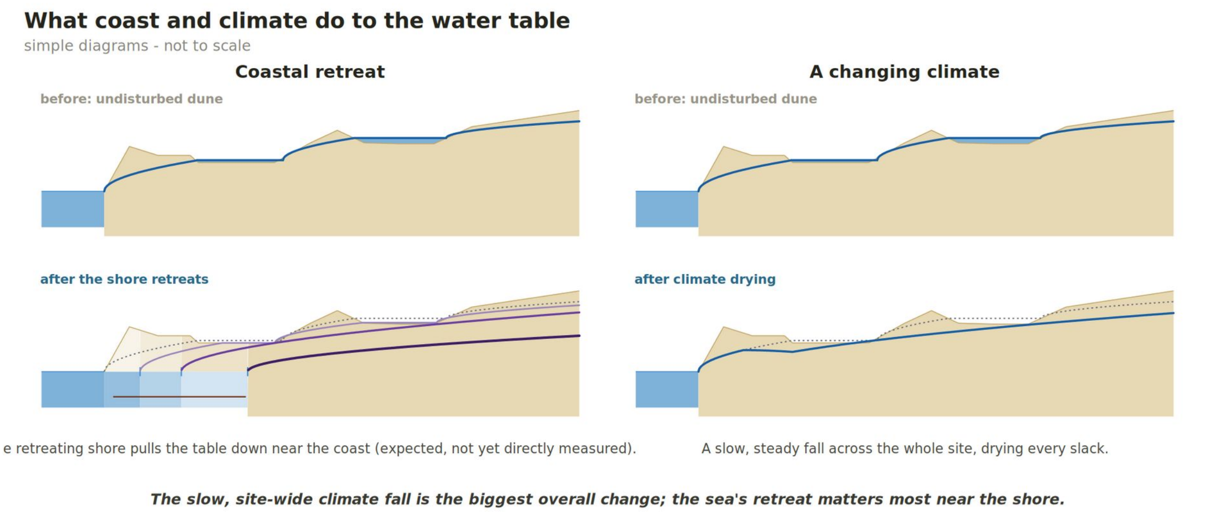

Figure 1 — Coast and climate. *The two dominant drivers, before and after: the retreating shore pulls the water table down near the coast, while a changing climate lowers it slowly across the whole site. The sea's retreat is the largest change we can measure, and it matters most near the shore; the site-wide climate fall acts everywhere, but its rate cannot be measured from this record. Schematic, not to scale.*

Management helps — but it is limited, and one tool can backfire

Clearing the forest. Felling the conifer plantation raised the *average* water table where the trees came out — by around **12 centimetres over the year** in our best estimate. But it did **not** lift the **summer low point** — and the summer low is the moment that actually decides whether a slack stays wet. So felling helps, modestly, but it does not fix the season that matters most.

Scraping slacks. Scraping — stripping a slack back down to the water table — reliably makes the *scraped hollow itself* wetter. But near the coast it works by the **same mechanism as erosion**: cutting back into the landward edge of a slack draws the water table down in the ground just around it. So a scrape can **dry its neighbours even as it wets itself**. Placement is everything.

Both interventions are **local** — they change the worked slack and its close neighbours, not the whole site — and both are **dwarfed by coastal erosion and by the warren-wide decline** (Figure 2). Management is not pointless; but it is a scalpel, not a lever on the whole Warren.

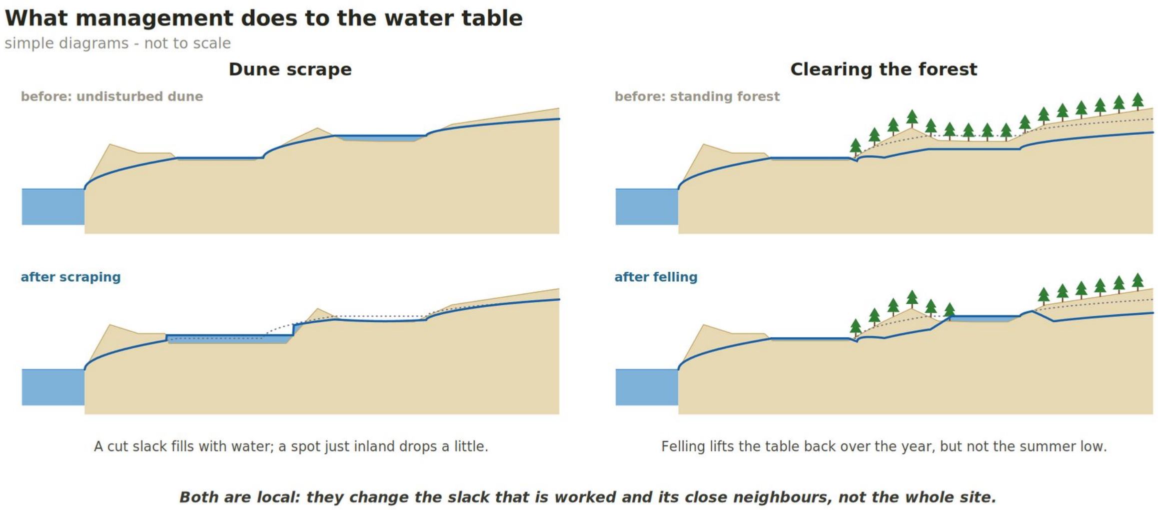

Figure 2 — What management does. *Dune scrape (left) and forest clearing (right), before and after. A cut slack fills with water but can draw down a spot just inland; felling lifts the table over the year but not the summer low. Both effects are local. Schematic, not to scale.*

The problem is getting harder to see

As the water table falls, the summer low increasingly drops **below the bottom of the measuring wells** — so the driest, most important moments are the hardest to capture. To stay honest, the study leans on the **spring record**, which is nearly complete. Spring is the seasonal high, so this is a deliberately **conservative** choice: if anything, it *understates* the drying.

The window is closing

The vulnerable slacks on the eastern side of the Warren have been flickering back and forth across the wet/dry threshold **since around 2022**. Ground that spends more and more of its time on the dry side of that line steadily loses its wet-slack character. The time to act on the ground that can still be saved is **now**, not later.

What this leaves for the people managing the dunes

The study hands the people who look after the Warren — and other dune systems like it — a set of practical tools:

- **An achievability map** — well by well, where wetter conditions are realistically within reach and where they are not, so effort goes where it can actually work.
- **Simple rainfall rules** — thresholds linking a season's rainfall to the water levels to expect, usable without specialist modelling.
- **A cheap, repeatable method** — the whole approach runs on hand-measured wells and free public weather data, so it can be kept going here and **transferred to other dune systems** facing the same pressures.

The bottom line

Newborough's slacks are under two large, mostly external pressures — a site-wide **climate** drawdown and a **coastal-erosion** front — that together outweigh what management can do. Management still matters: placed well, and judged against the **summer low rather than the annual average**, it can buy time and protect the best remaining ground. But the two big drivers set the direction of travel, and for the most vulnerable slacks the window is closing.

Further information

This summary draws on the full technical report:

> Hollingham, M. (2026). *Hydrogeological Dynamics, Behavioural Clustering and Management Intervention Analysis at Newborough Warren Coastal Sand Dune Aquifer, Wales.*

The report itself, together with the interactive tools, all the figures, and the underlying data and code, is available at the project website:

https://newbroman.github.io/Newborough_Hydrology/

About this study: an independent 21-year water-table monitoring programme at Newborough Warren SAC, Anglesey. The method relies on a network of 88 manually read dipwells and public climate data, and is designed to be low-cost, repeatable, and transferable to other dune systems.